The problem:

The **cabbage stem flea beetle** (*Psylliodes chrysocephala*), in German: Großer Rapserdfloh, is a major pest of oilseed rape (*Brassica napus*).
In the field it feeds on the young emerging cotyledons (=embryonic leaves found within a plant‘s seed) and small true leaves in autumn (up to 5 leave stage).
Symptoms:

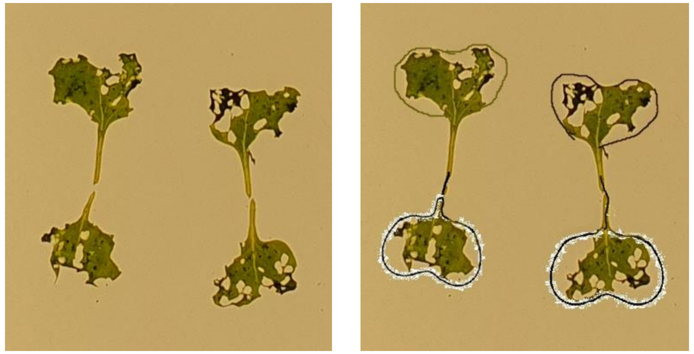
- **Shot-Holing:** Distinct, small, rounded circular holes on cotyledons and early true leaves.
- **Pitting:** Shallow bites and irregular indentations on young stems and developing leaves.

Contact: JUSTUS-LIEBIG-UNIVERSITÄT GIESSEN

christian.obermeier@agrar.uni-giessen.de
lennard.ehrig@agrar.uni-giessen.de

We aim to identify rapeseed cultivars which are resistant to feeding damage to use in plant breeding allowing agricultural cropping of oilseed rape with reduced insecticide treatments.
Damage quantification is currently done by labour, time-consuming and inaccurate visual assessment of digital images.
To speed this up we are looking for an automated image recognition and damage quantification pipeline.

Typical leaf feeding damage on young seedling of oilseed rape in the field



Typical leaf feeding damage on young seedlings (cotyledons) of oilseed rape under laboratory conditions (C. Posada-Vergara). Under severe damage estimating the total missing/eaten leaf area at the edges is difficult (right), especially from pictures taken in the field where leaves are somtimes twisted, upright and overlapping.

The data set:

- Digital pictures were taken from 300 or 400 plots (1,5 x 11 m) from 6 locations

Three pictures per plot,
300 or 400 plots per location
at one or two time-points

Field trials conducted in 2025 and damage caused by cabbage stem flea beetle (CSFB), creation of digital images

Partner	Location	Experiment	Field plot layout	Sowing date	1st CSFB Phenotyping	2nd CSFB Phenotyping
JLU	Groß-Gerau	Sulfur	4 x 100	8.9.2025	-	14.10. BBCH15 (1271)
JLU	Groß-Gerau	Insects	6 x 50	1.10.2025	21.10. BBCH10 (937)	10.11. BBCH13 (937)
NPZi	Malchow	Insects	10 x 30	29.8.2025	12.9. BBCH10 (900)	19.9. BBCH13 (900)
DSV	Asendorf	Insects	5 x 60	3.9.2025	15.-17.9. BBCH11 (910)	30.0./1.10.BBCH15 (900)
JLU	Weilburger Grenze	Insects	5 x 60	18.9.2025	7.10. BBCH10 (900)	20./24.10. BBCH15 (360+541)
JLU	Rauischholzhausen	Insects	3 x 100	2.9.2026	-	29.9. BBCH13 (900)

BBCH is the developmental stage of the plants (see link , e.g. BBCH11 means the two cotyledons and the first true leaf is fully developed)
In brackets: number of pictures taken, in total: 8946

Provided data set with folder name labelling schemes and detail information:

Partner	Location	1st CSFB Phenotyping	2nd CSFB Phenotyping	Zip folder of picture files, first part of file name identifies sampling date and BBCH in columns 3 and 4 (number of contained jpg files)	Folder size	Visual scoring finished, files (in main folder) and in folder RFSB-Phenotyping_training_set.zip
JLU	Groß-Gerau	-	14.10. BBCH15	2025_10_14_RSFB-Phenotyping_GG_Sulfur_JLU.zip (1271)	9.42 GB	-
JLU	Groß-Gerau	21.10. BBCH10	10.11. BBCH13	2025_10_21_RSFB-Phenotyping_GG1_JLU.zip (937) 2025_11_10_RSFB-Phenotyping_GG2_JLU.zip (937)	7.99 GB 7.52 GB	2025_10_21_RSFB-Phenotpyping_GG1_scores.csv (2 x 937) -
NPZi	Malchow	12.9. BBCH10	19.9. BBCH13	2025_09_12_RSFB_01_NPZi.zip (900) 2025_09_19_RSFB_02_NPZi.zip (900)	5.35 GB 4.48 GB	- -
DSV	Asendorf	15.-17.9. BBCH11	30.9./1.10. BBCH15	2025_09_15_Res4StRes_T1_DSV.zip (910) 2025_09_30_Res4StRes_T2_DSV.zip (900)	7.10 GB 7.54 GB	2025_09_15_Res4StRes_T1_DSV_scores.csv (910) -
JLU	Weilburger Grenze	7.10. BBCH10	20./24.10. BBCH15 BBCH15	2025_10_07_RSFB-Phenotyping_WG1_JLU.zip (900) 2025_10_20_RSFB-Phenotyping_WG2_FirstHalf_JLU.zip (360) 2025_10_24_RSFB-Phenotyping_WG2_SecondHalf_JLU.zip (541)	6.72 GB 2.85 GB 4.21 GB	2025_10_07_RSFB-Phenotyping_WG1_JLU_scores.csv (900) - -
JLU	Rauischholz-hausen	-	29.9. BBCH13	2025_09_29_RSFB-Phenotyping_RHH1_JLU.zip (900)	5.53 GB	-
JLU/ GAU	Scoring data	Only BBCH10-11	-	Three csv-files with scoring data (see column ‚Visual scoring finished…') plus a folder with 470 pictures which contains the most similar/reliable independent scorings from BBCH10 at 21.10. in GG (less than 5% difference between GAU and JLU scoring) which might be used as calibration data set	3.7 GB	Folder RFSB-Phenotyping_training_set.zip (470, subset from 937 pictures)

Note: Visual scoring is only finished for part of the data (2747 pictures). For 937 pictures scoring was done twice independently by GAU and JLU. Some scorings were quite different. However, for 470 pictures the (average) scores per picture were similar (less than 5% difference). We suggest to use these pictures as a calibration set. QR codes in the pictures are identical within 3 pictures taken from 3 different positions within the same plot (the same genotype) and QR codes are also identical for the same plots and different sampling time points (1st and 2nd CSFB phenotyping), only file *.jpg names are unique. JLU = Justus Liebig University, GAU = Georg August University

Link for downloading folder (65 GB) with image data organized in 12 subfolders as described in table above from next.Hessenbox: https://next.hessenbox.de/index.php/s/G82rsMRSMpzHapJ (password required)

The data set in more detail:

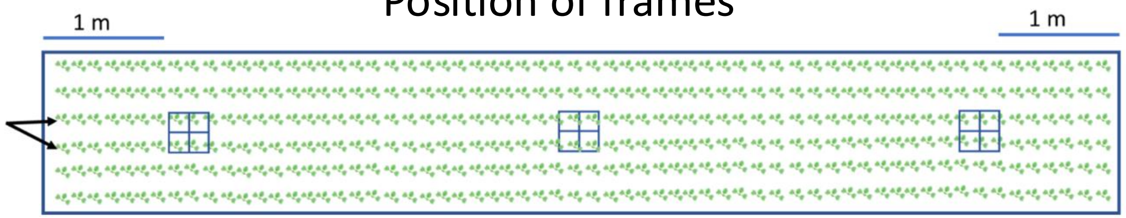
- Digital pictures were taken by attaching a mobile phone with camera to a frame covering 0.1 square metres
- From each plot 3 pictures were taken in different parts (position of frames)
- The frame is used for orientation, for fixing the camera distance and dispalying a QR code within the picture which can be used for identification of the genotype/variety by scanning the picture from a computer display (or automated software, always 3 pictures with the same QR code at each time-point of sampling)

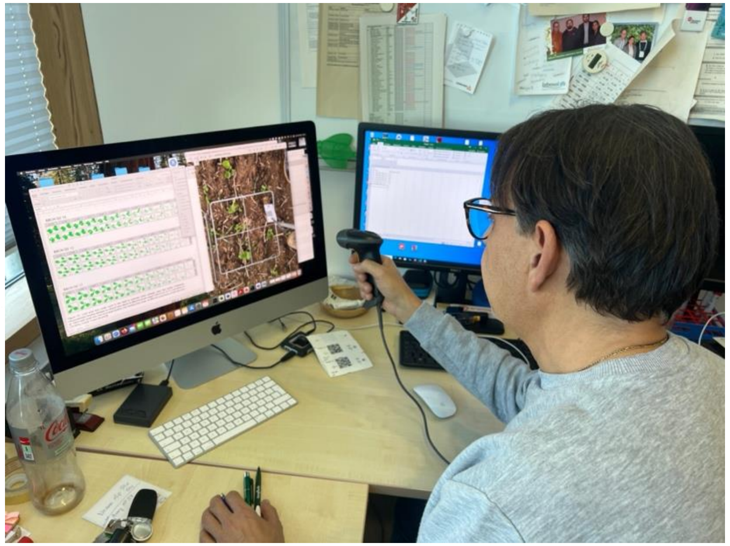
Assessment of feeding damage caused by the cabbage stem flea beetle using image data with the help of the ‚Göttinger Zähl- und Schätzrahmen‘

Position of frames

Plot: 1,5 x 11 m

Middle rows





Manual analysis of the data set :

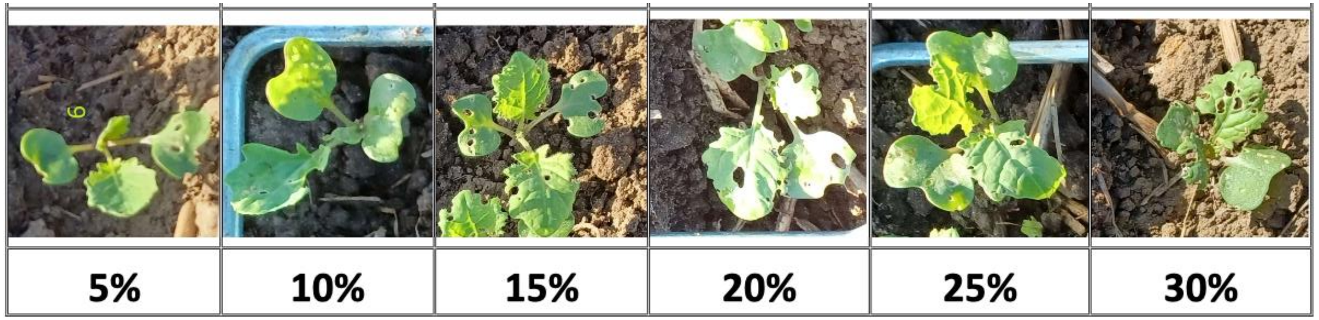
- The images were displayed on a computer screen, the barcode was scanned from the screen and a trained student was estimating the percent leaf damage (shot holes) for 10 plants one by one, or by giving an average across 10 plants and entering it in Excel.

- the estimation was based on comparison with visual damages scales deveolped by an entomologist (C. Posada-Vergrara) for BBCH10-11 and BBCH 12 (see pictures on the right).

- The problem is, that this is highly subjective. This can be seen on the next slide: for one location the scoring for the same 900 pictures was done independently at Georg August University (GAU) and Justus Liebig University (JLU) by different persons. Correlation is low: r=0.549.

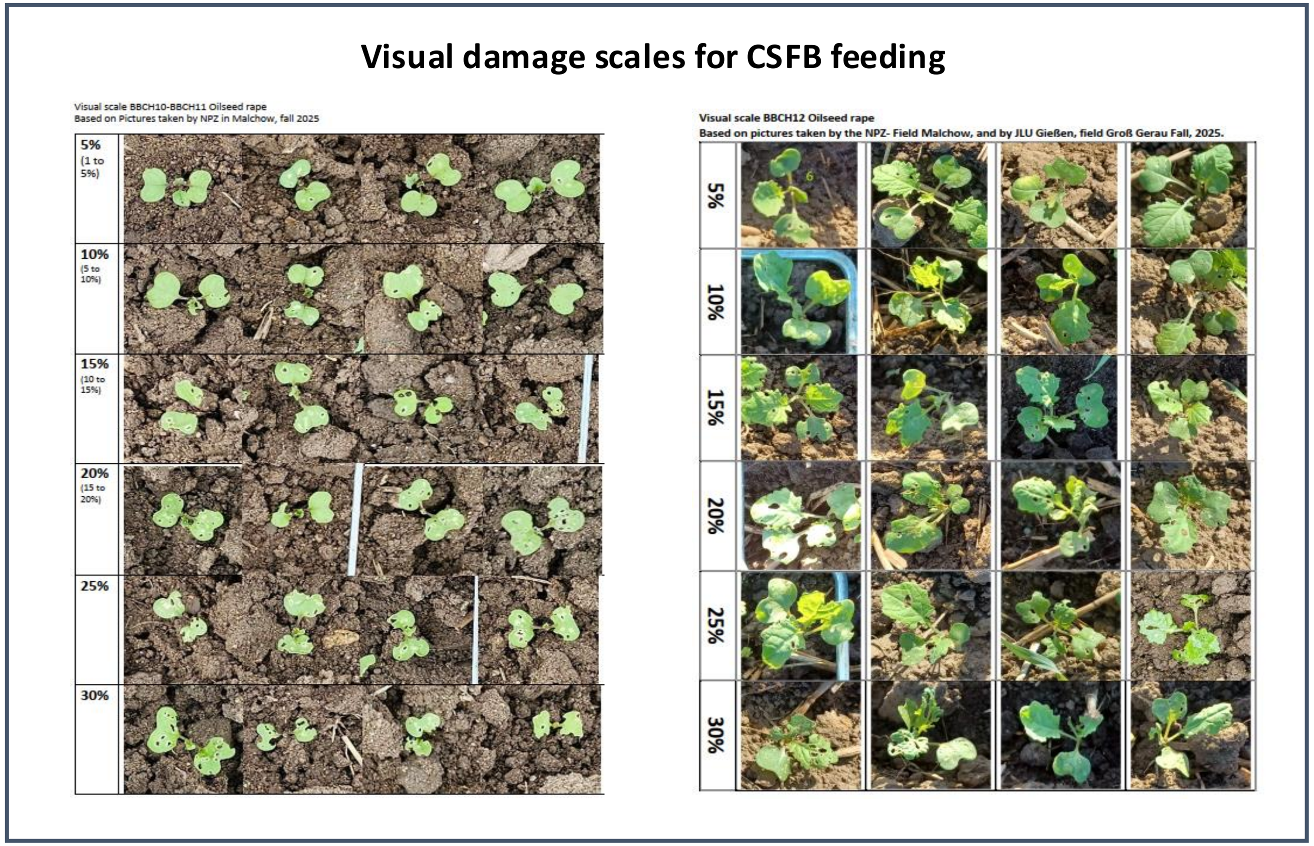
- Thus, it is not so easy to provide ‚ground thruth‘ data for callibration of an automated image recognition tool.

- We are suggesting to use a set of 470 representative pictures where different persons gave consistent scores (less than 5% difference).



BBCH10-11

BBCH12



Correlation between independent picture analyses

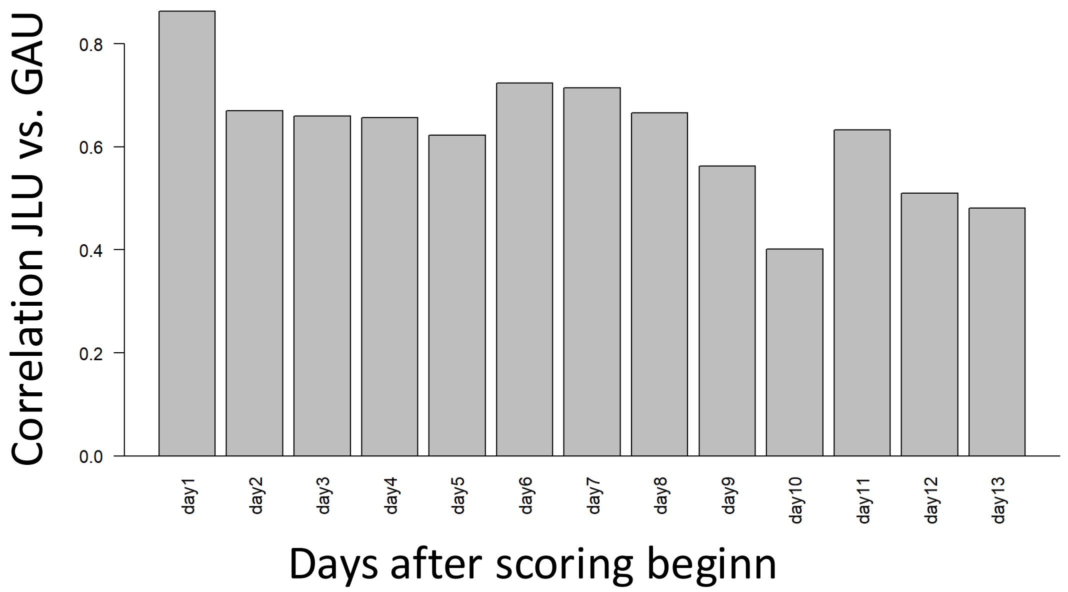
JUSTUS-LIEBIG-UNIVERSITÄT GIESSEN

UNIVERSITÄT GÖTTINGEN

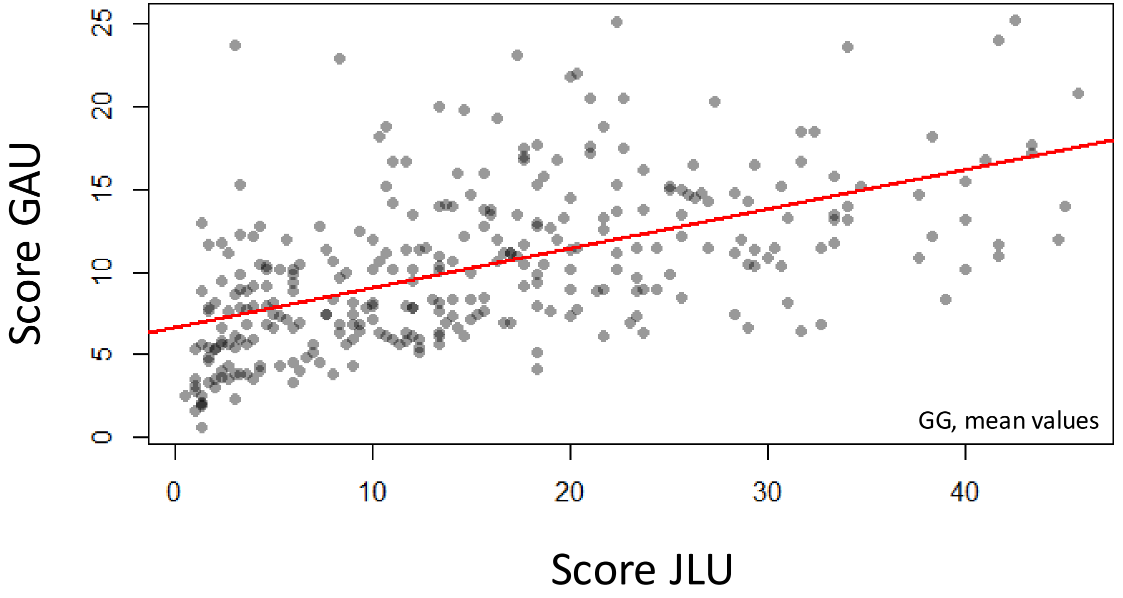
Lennard Roscher-Ehrig

Cabbage stem flea beetle, adults

Correlation for visual phenotyping between different trained students:
0.84 to 0.55 - Need for improvement



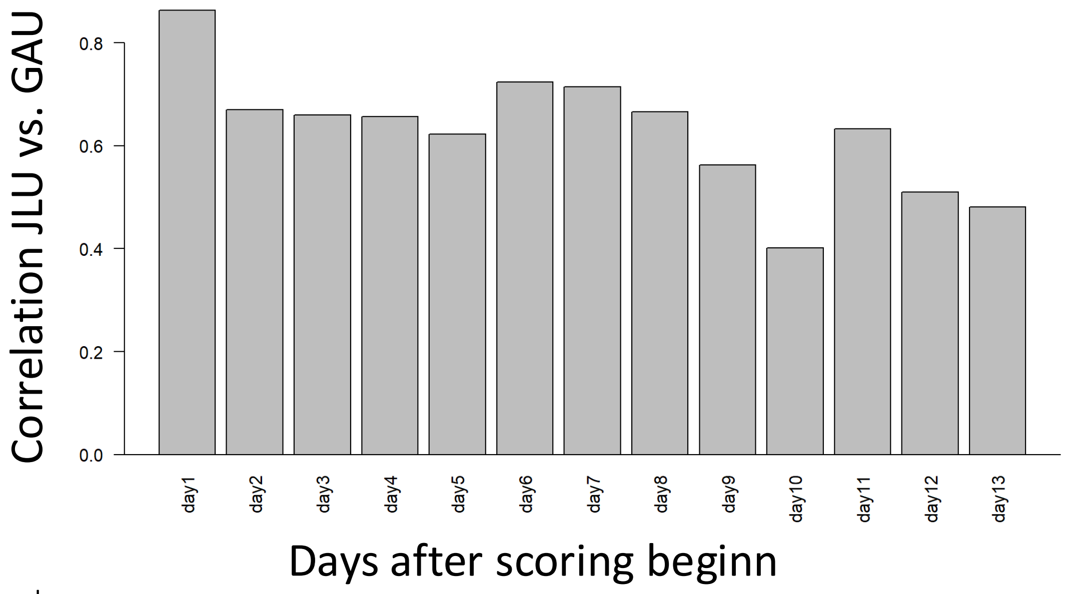
Correlation r=0.549 Groß-Gerau (n=900)



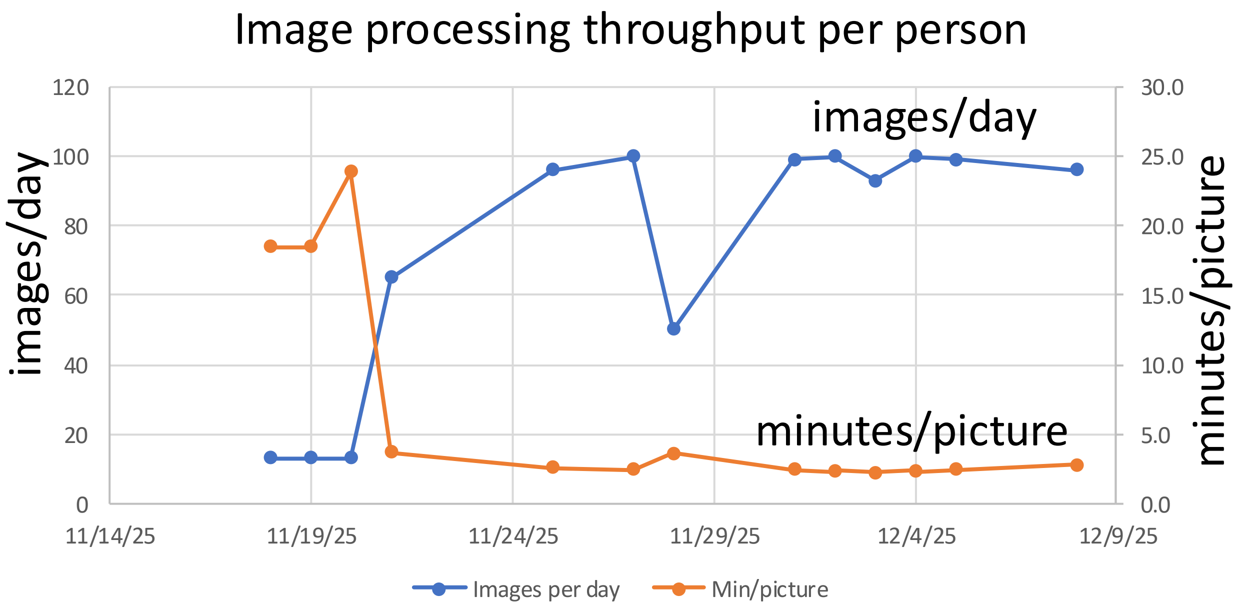
Correlation between independent picture analyses

Correlation for visual phenotyping between different trained students:
0.84 to 0.55 - Need for improvement
Conflict between accuracy and speed

for 8400 pictures visual scoring by 1 person and 20 minutes per picture would require 70 weeks (1,5 years) with a 40 h week, if scoring can be done in 2 minutes per picture this would still require 2 month per person



GG, mean values



Note: on day 1 directly after training correlation was highest with 0.84

Two example pictures from very young plants (BBCH10 to 12 = only two cotyledons, or two cotyledons plus first and second leaf visible) and from older plants (BBCH 15 = up to 5 true large and partially overlapping leaves are visible).

<u>Required values from machine learning picture processing output for evaluating genotypes for resistance against leaf feeding:</u>

- number of shot holes per plant.

- number of yellow spots (feeding without producing holes) per plant.

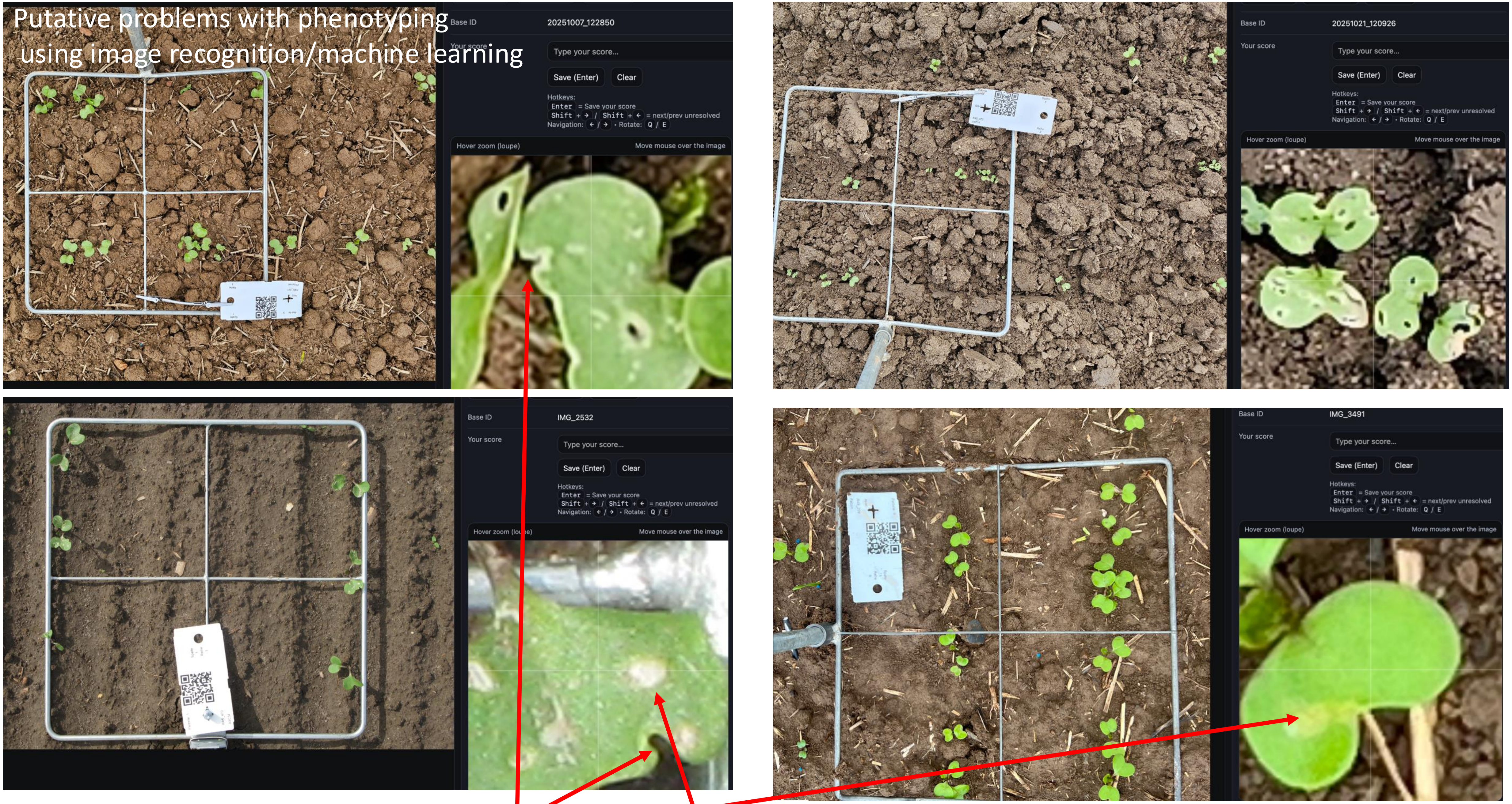
- area of holes and spots in % = percentage of leaf area with different types of damage per plant (including missing area at the edges of leaves).

- number of single plants on the picture and within the metal frame.

- average number of leaves per plant (2 = just cotyledons or 3, 4, cotyledons, plus first/second true leaf) for first time point of picture sampling.

Plants from different time points (BBCH10-11 vs. BBCH 13-15) look very different: much larger at BBCH15 with more bigger/overlapping leaves with irregular holes also at the edges of leaves. For BBCH10 the problem might be that the plants and the holes/spots are very small on single plants. Probably data processing needs other requirements for different time-points.

Putative problems with phenotyping using image recognition/machine learning



If the beetle damaged the edges of leaves this is difficult to distinguish from natural growth.

Here the beetle feeding did not produce holes, but just yellow patches. Ideally, this needs to be distinguished from true hole symptomes.

Background info on associated research project:

Contact:
Christian Obermeier
christian.obermeier@agrar.uni-giessen.de, 0641-99 37426

Lennard Roscher-Ehrig
lennard.ehrig@agrar.uni-giessen.de, 0641-99 37432

Department of Plant Breeding
Justus Liebig University Gießen
plantbreeding-giessen.de
res4stres.de